

Jiayang Shang (Dylan)

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EDUCATION

University of New South Wales

Sydney, Australia

Master of Information Technology (Excellent graduate) WAM: 82

Feb. 2023 – Sep. 2024

Core Courses: Principles of Programming (HD), Data Structures and Algorithms (HD), Database Systems (HD), Artificial Intelligence (DN), Computer Networks (HD), Software Construction (HD)

Jiangsu University of Technology

Jiangsu, China

Bachelor of Internet of things

Sep. 2018 – Jun. 2022

Core Courses: Digital/Analog circuits, Mobile application development, Short-distance wireless technology, Embedded software development, Microprocessor Application, etc.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, C#, SQL, JavaScript, TypeScript, PHP, Shell, Java

Frontend Frameworks & Styling: React, Vue, Next.js, Tailwind, HTML, CSS

Backend Frameworks: Node.js, Express.js, Flask, FastAPI, Laravel, WordPress, Django

APIs & Testing: GraphQL, REST, Jest, Cypress

Libraries & Tools: OpenCV, Selenium, QT, WebAssembly, Jira

Cloud & DevOps: Git, Docker, CI/CD, Google Cloud Platform, AWS, Azure

EXPERIENCE

Full Stack Software Engineer

Apr. 2025 – Present

Golden Matrix Group Inc.

Sydney, NSW, Australia

- Designed and developed full-stack applications using Java, Laravel, MySQL, Vue, React, Redis, Docker, and AWS, with CI/CD pipelines for automated deployment.
- Built and optimized a high-concurrency reporting system, significantly improving performance, stability, and scalability under heavy traffic conditions.
- Led the migration of a clustered system from a traditional server-based architecture to AWS ECS and elastic containerized infrastructure, improving resource utilization and deployment efficiency.
- Integrated third-party platforms including Braze and Revolut, enabling customer engagement automation and secure payment workflows across the platform.

Software Engineer

March. 2025 – April. 2025

Aquion Pty Ltd

Sydney, New South Wales, Australia

- Develop software with Jira, Docker, Laravel, Mysql, Vue.js, OpenAI API, CI/CD
- Developed a vendor portal enabling primary customers to review 6,500+ vendors and products, integrated with ERP for real-time data.
- Enhanced a custom CRM and Data Lake to streamline customer data capture and integrate with the internal quoting system.
- Integrated OpenAI to classify vendors, find their website URLs, and update information in the CRM system.

Software Engineer

Dec. 2021 – Jun. 2022

Youzhi Automation Technology · Internship

Jiangsu, China

- Developing industrial vision software with C++, C#, OpenCV and QT
- Implementing and optimizing the vision algorithms and improving the interactivity of software
- Engaged in performance benchmarking of software systems, systematically evaluating metrics, and analyzing results

SELECTED PROJECTS

Kùzu-Wasm [Link] | *C++, Javascript, React, Webassembly, Apache Arrow, Cypher* Feb. 2024 – Dec. 2024

- First public Graph Database powered by WebAssembly
- Developed during the research program at the UNSW under the supervision of Dr. Zhengyi Yang
- Designed to provide fast server-less analytical processing for the graphs on the Web
- Features: Extremely fast, Highly Scalable, Serverless, Embeddable, Cypher Support
- Collaborated with Alibaba's GraphScope team and integrated the solution into the portal product.

Industrial Vision Processing Program [Link] | *C++, C#, Qt, OpenCV, PCL* Dec. 2021 – Jun. 2022

- Developed a machine vision software named AIV with C++, which was used in industrial product quality testing
- Using OpenCV to accomplish 2D/3D features including template matching, line/circle/plane finding, etc.
- Implemented UI functions including step-by-step execution, scripting, debugging, 2D/3D effect preview, etc.

Campus Guide Program [Link] | *PHP, Python, MySQL, React* Jul. 2018 – Present

- Built a WeChat program with React which provides campus guide including encyclopedia, news, personal class schedule, and learning materials for all staff and students
- Implemented Restful API interface with Laravel framework and used Python to collect course information
- The program has been maintained since 2018, and the number of users is over 10,000

PUBLICATIONS

Pine Script v5 User Manual Chinese Version: Script Manual for TradingView, published on Feb. 2022 on GitHub [Github] [Online Book]

Kùzu-Wasm: High-performance In-browser Graph Analysis: Under the supervision of Dr. Zhengyi Yang and published on Aug. 2024 during the research program at UNSW [Link]